VCH Lab- Case Study Analysis

Structural Diagnosis of Documentation, Traceability, and Reproducibility in the VCH Lab

This analysis examines each track individually. The purpose is not only to identify what research outputs remain, but to explain what their presence or absence reveals about how the research process actually unfolded. For each case, the analysis reconstructs how data was generated, how decisions were made, and how knowledge was preserved or lost. This allows the analysis to move beyond description toward a causal explanation of why reproducibility breaks down. Each finding is therefore structured to make the full reasoning process explicit, ensuring that the connection between observed evidence and analytical conclusion is fully transparent. In other words, the goal is not simply to say that something is missing, but to explain what is missing, where it disappeared, why that happened, and what that means for research quality.

The analysis is structured around five interrelated dimensions: data traceability, process documentation, decision traceability, knowledge preservation, and reproducibility. These dimensions reflect established requirements for transparent and reusable research. The FAIR principles define that research outputs must be findable, accessible, interoperable, and reusable, meaning that both data and the context needed to interpret it must be clearly structured and retrievable (Wilkinson et al., 2016). In plain terms, this means that research material should not only exist, but should also be easy to locate, understand, and use again later. Provenance theory requires that the full lineage of data, including transformations, decisions, and intermediate steps, is preserved so that results can be independently reconstructed (Buneman et al., 2001). Put simply, provenance means keeping a visible record of how the research moved from starting data to final conclusions. Knowledge conversion theory explains that when reasoning remains tacit rather than being externalised into explicit documentation, it cannot be shared or reused, leading to discontinuity between projects (Nonaka & Takeuchi, 1995). Here, tacit means knowledge that remains in people’s heads, conversations, and personal understanding instead of being written down in a visible and structured form. Together, these perspectives establish that reproducibility depends on the continuous visibility of how knowledge is produced, not only on the existence of final outputs.

Across all cases, meaningful research activity is consistently present. The central issue is not the absence of work, but the absence of a system that ensures that work is captured, connected, and preserved in a way that supports reuse. This point is important because it shifts the analysis away from blaming individual students and toward understanding the structural conditions that shape how documentation happens in practice.

Bakery Track

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Research outputs and deliverables

- Audio recording of stakeholder interview (Bbrood)
- Final presentation slides (summary of findings)

Missing research outputs and deliverables

- Desk research notes used to prepare the interview
- Interview preparation notes or question design
- Informal team notes created during the project

From the available materials, it is clear that primary research activity did take place in the Bakery track. A recorded stakeholder interview is available, confirming that empirical data were collected. However, this data exists in isolation, without a transcript, coded analysis, or structured interpretation linked to it. This indicates that, while the project successfully gathered real input from a stakeholder, the material explaining how that input was interpreted and applied is missing.

Reconstructing how the work likely unfolded suggests that the interview formed part of a broader research process that included preparation, question design, and interpretation of responses. The student would have translated spoken input into insights, identified patterns, and used these to inform conclusions. Yet none of these intermediate steps were formally documented. As a result, although both the starting point and final outcome of the research process are visible, the transformation between them is missing. This missing middle section is critical, as it is the part that reveals how the team moved from raw conversation to structured meaning.

The underlying cause of this breakdown lies in the absence of embedded process documentation during execution. Preparatory work, interview design decisions, and analytical steps were treated as part of the working process rather than as elements requiring explicit recording. This caused the reasoning to remain internal and situational rather than externalised and durable. In practical terms, the thinking did occur, but it remained within the workflow instead of being translated into a clear and accessible record that others could follow.

Knowledge conversion theory explains this mechanism by showing that tacit reasoning must be converted into explicit form to become shareable and reusable (Nonaka & Takeuchi, 1995). In this context, tacit reasoning refers to the team's internal understanding of what the interview meant, why certain questions were asked, and how conclusions were formed. Because this conversion did not occur, the reasoning process disappeared once the project ended. At the same time, provenance theory clarifies that the sequence of transformations connecting raw data to conclusions must be preserved to allow reconstruction (Buneman et al., 2001). In this case, that sequence is incomplete. Put simply, there is no visible trail showing how the interview was translated into findings.

As a result, reproducibility breaks down. Although raw data and final outputs are available, it is not possible to determine how specific interpretations were made, what analytical steps were taken, or how conclusions were derived. From a FAIR perspective, the data is not reusable because the contextual information required to interpret it is missing (Wilkinson et al., 2016). This means that someone else may be able to access the file itself, but still cannot meaningfully reuse it because they would not understand how it was interpreted by the original team. This shows that without continuous documentation, valid data cannot be transformed into reproducible knowledge.

Cacao Track

Research outputs and deliverables

- Final written guide on ethical cacao sourcing and sustainability
- Collection of referenced sources (reports, policy documents, academic literature)
- Draft versions of the guide

Missing research outputs and deliverables

- Notes comparing sources
- Early drafts and discarded content
- Internal reasoning behind source selection

Looking at the available materials, the Cacao track presents a well-structured final guide supported by multiple sources. This indicates that substantial research and synthesis took place, and that the student engaged with a range of materials to construct the final output. At first glance, this creates the impression of a more complete and rigorous research process than in the Bakery case, as the output appears polished, coherent, and grounded in existing literature.

Tracing how the research was likely carried out shows that the student reviewed multiple sources, selected relevant information, compared perspectives, and combined insights into a coherent narrative. This process likely involved filtering sources, resolving inconsistencies, and deciding how different pieces of information should be integrated. However, the specific steps through which this synthesis occurred are not visible. There is no mapping between individual sources and specific claims in the final guide. As a result, the guide presents the outcome of the synthesis, but not the pathway that produced it.

The core driver behind this issue is the absence of structured documentation of analytical steps during the synthesis process. While sources were collected and used, the reasoning that connected them to the final output was not externalised. Decisions regarding which sources to prioritise, how to interpret them, and how to resolve conflicting information were made during the process but not recorded. This matters because research synthesis is not only about gathering sources, but also about making informed choices between them, and those choices directly shape the final argument.

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Provenance theory explains that without documentation of how inputs are transformed into outputs, the analytical pathway cannot be reconstructed (Buneman et al., 2001). In this context, provenance refers to the visible chain linking source material to final claims. Knowledge conversion theory further clarifies that when reasoning remains tacit, it cannot be validated or reused by others (Nonaka & Takeuchi, 1995). Here, tacit refers to the logic of selection, comparison, and interpretation remaining within the student's working process instead of becoming part of the documented research record. Although the output appears coherent, the absence of explicit reasoning limits its transparency. This becomes critical because readers can see what was concluded, but not how the conclusion was methodologically constructed.

As a result, the research remains understandable but not reproducible. It is not possible to trace how specific claims were derived from specific sources, or to evaluate the validity of the synthesis process. From a FAIR perspective, the output lacks the structured context required for reuse (Wilkinson et al., 2016). In practical terms, someone else may be able to read the guide, but would not be able to confidently continue the research, test the claims, or reconstruct the same synthesis process. This highlights that well-developed outputs do not guarantee research validity if the process that produced them is not visible.

Textile Track

Research outputs and deliverables

- Visual research boards (e.g., Canva, Miro)
- Presentations and visual mappings of insights

Missing research outputs and deliverables

- Exported files or archived versions of boards
- Written analyses supporting visual outputs
- Source references and traceable links
- Documentation of analytical steps

Based on the available outputs, the Textile track presents research work primarily in the form of visual boards and presentations. These outputs organise insights into categories and themes, indicating that analysis and structuring of information took place during the project. This suggests that the team did not simply produce visuals, but used them as a means of organising and interpreting data.

Examining how the work was likely carried out indicates that the student collected data, interpreted it, and translated it into visual representations. Insights were grouped, relationships were identified, and conclusions were embedded within the structure of the boards. However, the connection between these visual elements and their underlying sources is not documented. There are no explicit references, source links, or explanations showing how specific insights were derived. As a result, the visuals demonstrate that analytical thinking occurred, but do not reveal where that thinking originated or how it was justified.

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This breakdown emerges from a reliance on tools that prioritise visual organisation over traceable documentation. The research process became embedded within the platform environment, meaning that insights were captured as outputs rather than as part of a documented reasoning process. Because documentation was not externalised beyond the tool, the underlying logic remained implicit. In practical terms, the work was stored in a format that displays the outcome of the thinking, but not the reasoning behind it.

From a theoretical perspective, FAIR principles require that data be both findable and interpretable, which depends on clear references to sources and structured metadata (Wilkinson et al., 2016). In this context, metadata refers to the additional information that explains what something is, where it came from, and how it should be understood. Lifecycle perspectives further show that documentation must occur throughout each stage of the research process, rather than only at the end (OECD, 2021). In this case, documentation was effectively replaced by visual output. This becomes critical because a visually clear or engaging result is not equivalent to a complete and traceable research record.

As a result, both knowledge preservation and reproducibility break down. If access to the platform is lost, the research disappears entirely. Even when access is maintained, the absence of explicit references prevents reconstruction of how conclusions were reached. This means that the project is fragile in two ways: it can be lost technically, and it can remain unclear analytically. This shows that tool-based outputs without structured documentation cannot function as stable and reliable research records.

CRM Track

Research outputs and deliverables

- Presentations and slide decks
- System maps and conceptual models
- Draft documents and working files

Missing research outputs and deliverables

- Interview transcripts
- Structured documentation of analytical steps
- Version history showing development over time
- Source-to-claim mappings
- Centralised and structured repository of materials

A review of the available materials shows that the CRM track contains a large volume of outputs, including presentations, system maps, and draft documents. These materials indicate extensive research activity and strong engagement with the topic. Compared to the other tracks, this case creates the strongest impression of a rich and active research process, as there is a substantial amount of visible work.

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Tracing how the project likely developed suggests that it involved multiple iterations, stakeholder interactions, and evolving problem definitions. The student appears to have gathered input from stakeholders, analysed policy or system-level information, and adjusted the direction of the project over time. However, the connections between these stages are not documented. It is not possible to determine which inputs informed which outputs, or how the project progressed from one stage to the next. As a result, although the project appears substantial, its internal development remains difficult to follow.

This breakdown can be traced to a lack of alignment between research activity and documentation requirements. The system in which the project was conducted prioritised solution development and stakeholder engagement, but did not require structured recording of how decisions were made or how evidence was used. As a result, documentation remained fragmented and incomplete. This becomes significant because the CRM case shows that even high levels of effort do not automatically produce high traceability. Without a structure that enforces connections between steps, outputs can accumulate without becoming methodologically transparent.

Work System Theory explains that outcomes are shaped by the interaction between tasks, tools, roles, and evaluation criteria (Alter, 2013). In this case, the system did not require traceability, and therefore, traceability did not emerge as a consistent practice. In practical terms, the way the project was organised shaped what students prioritised. If the system mainly rewards producing solutions, then students will focus on producing solutions. Knowledge conversion theory further shows that when reasoning remains unrecorded, it cannot be preserved beyond the immediate context (Nonaka & Takeuchi, 1995). This means that the work may be understandable to those involved during the project, but becomes difficult to reconstruct afterward.

As a result, despite the volume of outputs, the research is not reproducible. It is not possible to reconstruct how decisions were made, how evidence informed those decisions, or how the project reached its final form. This demonstrates that output quantity does not compensate for missing traceability. A large collection of materials can still fail to function as a coherent research record if the relationships between them are not preserved.

Organizational Development Track

Research outputs and deliverables

- Orga master deck presentation
- Photos taken during the final presentation

Missing research outputs and deliverables

- No research documentation
- No project files
- No analytical outputs

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- No stored datasets
- No documentation of student work processes
- No validation or methodological records

An examination of the available materials shows that the Organizational Development track reflects the broader structure within which all projects operate. The programme defines goals and encourages flexibility, but does not specify how research processes should be documented. This makes the track particularly important, as it not only reveals a project-level issue but also clarifies the conditions under which the other projects were conducted.

Considering how this structure influences project execution shows that students are left to determine their own documentation practices. Some may choose to record aspects of their work, while others may not, leading to inconsistency across projects. Because no standards are enforced, documentation becomes optional rather than an integral part of the research process. In practice, this means that the quality of the research record depends heavily on individual behaviour instead of being supported by a shared system.

This pattern can be explained by a lack of socio-technical alignment. Documentation is not required by the system, not supported by tools, and not reinforced through evaluation criteria. As a result, it does not become a stable or consistent behaviour across the lab. Socio-technical alignment refers to the need for both the technical environment, such as tools and systems, and the social environment, such as expectations and routines, to support the same behaviour. In this case, that alignment is absent.

Socio-technical systems theory explains that consistent practices emerge only when technical structures and social expectations are aligned (Trist, 1981). Institutional benchmarking further shows that mature research environments define documentation standards, metadata requirements, and storage systems at the outset of projects (NWO, 2020; Science Europe, 2022). In this context, metadata refers to the structured contextual information that allows data and outputs to be identified, interpreted, and reused. This comparison is important because it shows that the issue within the VCH Lab is not simply normal variation, but an indication of lower research process maturity.

As a result, a systemic failure of reproducibility occurs. Because documentation is not embedded at the system level, the same issues appear across all projects. This shows that the problem is structural rather than individual. In other words, the recurring weaknesses are not random, but are produced by the way the environment is currently organised.

Cross Case Analysis: System Level Structural Failure

Across all five cases, four recurring failure mechanisms emerge. These mechanisms are interconnected and reinforce one another, forming a system that consistently produces the same outcomes. This means the cases should not be read as separate accidents. They should be read as different expressions of the same deeper problem.

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These patterns are consistently observable across the Bakery, CRM, Cacao, Textile, and Orga tracks, although they manifest in slightly different forms depending on the project structure and available materials. Despite these variations, the underlying mechanisms remain the same, indicating that the issue is structural rather than case-specific.

First, documentation is not embedded within the research workflow. Instead, it is concentrated near submission, meaning that intermediate steps are not recorded. This pattern is visible across multiple cases. In the Bakery track, interview preparation and analytical steps are not preserved. In the Cacao track, synthesis and reasoning steps are missing between data collection and the final output. In the CRM track, iterative development and intermediate decisions are not traceable in the repository. Reconstructing this pattern shows that once these steps are lost, they cannot be recovered. Lifecycle theory explains that documentation must occur continuously to preserve reasoning (OECD, 2021). Here, lifecycle means the full sequence of stages through which a project develops, from the start of the work through analysis and final output. This matters because if documentation only happens at the end, then the most important parts of the reasoning process have already passed and cannot be recreated accurately later.

Second, decision-making remains tacit. Because documentation is not required, decisions are made through discussion but not recorded. This pattern appears consistently across cases where interviews indicate that reasoning was actively discussed during the project but not preserved afterward. For example, in both the Bakery and Cacao tracks, participants describe analytical reasoning processes that are not visible in the stored materials. This leads to a situation where key reasoning is known during the project but disappears afterward. Knowledge conversion theory explains that without externalisation, tacit knowledge cannot be reused (Nonaka & Takeuchi, 1995). Here, tacit decision making means that decisions are understood by the people involved while they are working, but the reasoning stays in conversation, memory, or shared intuition instead of being written down. This matters because decisions shape the final outcome, and if those decisions are not visible, the project cannot be properly understood.

Third, provenance chains are broken. Without recorded decisions and intermediate steps, it is not possible to trace how data is transformed into conclusions. This pattern is observable across all cases where final outputs exist, but the transformation process is not reconstructable. For example, across the CRM and Textile tracks, outputs can be accessed, but the analytical pathway that produced those outputs cannot be followed step by step. Provenance theory defines this traceability as essential for reconstructability (Buneman et al., 2001). In plain language, provenance means the visible chain that shows how the research moved from starting data, through analysis and reasoning, to conclusions. When that chain is broken, the final output may still exist, but the path that produced it can no longer be followed. That matters because it makes it impossible for another person to independently check or rebuild the work.

Fourth, reuse fails under FAIR principles. Because data lacks context, structure, and traceable reasoning, it cannot be reused effectively (Wilkinson et al., 2016). This pattern is consistent across all analysed cases, where materials are accessible but lack the necessary explanatory depth to support reuse. In simple terms, the research may still be visible, but it is not usable in a practical research sense because later students or researchers do not have enough explanation to work with it confidently. This is why reuse is not only about access to files. It is also about access to the logic that makes those files meaningful.

These mechanisms reinforce each other. The absence of documentation leads to tacit decision-making. Tacit decision-making breaks provenance chains. Broken provenance prevents reuse. The inability to reuse research prevents the accumulation of institutional knowledge, reinforcing the initial absence of documentation. This reinforcing loop is visible across all five cases, where similar breakdowns occur regardless of project topic or team composition.

This circular pattern is important because it shows that the problem is self-reinforcing. Once the system starts producing incomplete records, those incomplete records make it harder for later groups to improve the situation. This system-level pattern persists across cohorts because it is embedded in the structure of the lab rather than in individual behaviour. Each new project begins under the same conditions, leading to the repetition of the same issues. That is why the problem continues even when students are motivated and do meaningful work.

These findings directly answer the main research question by demonstrating that the lack of reproducibility within the VCH Lab is not caused by isolated project-level failures, but by consistent and recurring structural conditions that affect how research is conducted, documented, and preserved.

The conclusion is clear. The VCH Lab functions as a project-based learning environment rather than a reproducible research system. While it produces valuable outputs, it does not consistently preserve the processes, decisions, and data required for those outputs to be reused. This prevents the lab from functioning as a cumulative knowledge system and limits its long-term impact. A structured research framework is therefore not an enhancement, but a necessary condition for transforming isolated project work into reproducible and reusable knowledge.

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